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Effect of Pulsed Photon Treatment on the Mechanical Properties of Semiconductor Thermoelectric Legs, Based on Bi_2Te_3–Bi_2Se_3 Solid Solutions, and the Adhesion of Switching Layers

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Abstract—We carry out comparative studies of the phase composition, morphology, and hardness of semiconductor legs based on the *n*-type solid solution Bi_2Te_3–Bi_2Se_3, obtained by hot pressing, after surface modification (mechanical processing and pulsed photon treatment (PPT) with incoherent light). Using shear tests, we determine the adhesion of switching and barrier Mo–Ni layers on the modified surfaces of semiconductor legs. Pulsed photon treatment stimulates local recrystallization of the imperfect layer near the surface of samples of the Bi_2Te_3–Bi_2Se_3 solid solution to a depth of 100–200 nm, which increases the hardness of the surface layers. It is shown that the mechanical polishing and subsequent pulsed photon treatment of thermoelectric legs increases the adhesion of the switching and barrier Mo–Ni layers by 3–4 times, which can contribute to the efficient and stable operation of a thermoelectric generator battery.

Keywords: thermoelectric generator, Bi_2Te_3–Bi_2Se_3, pulse photon treatment, surface modification, mechanical properties

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INTRODUCTION

Increasing the efficiency of thermoelectric devices, along with the search for new materials, includes the development of new technologies for the implementation of the metal–semiconductor switching connection. Today, the main technological problem in the field of switching is the creation of contacts that ensure high adhesion with low contact resistance, which are a necessary condition for maintaining the reliability and durability of thermoelectric generator batteries [1].

The value of adhesion depends on the thickness, composition, and structure of the damaged layer, as well as the morphology and surface topography. For example, electrochemical polishing, ultrasonic treatment, drying in toluene or isopropyl alcohol vapor, and annealing in vacuum increased the metallization adhesion for solid solutions of bismuth chalcogenides, obtained by zone recrystallization, to 19 N/mm^2 and ensured a contact resistance of no more than 10^{-10} Ohm/m^2 [2]. Feng et al. [3] demonstrated that the combined application of various types and modes of polishing could reduce the surface roughness of a semiconductor leg to 2 nm. However, creating smooth and chemically clean surfaces of semiconductor legs is not in itself a sufficient condition for the reliable, durable, and reproducible electrical characteristics of contacts. The low roughness achieved in [3] is expedient, first of all, in microelectronics, while in the legs of powerful thermogenerator batteries, on the contrary, a moderately developed surface relief is preferable, because it increases the adhesion of the barrier layer and improves the *Q* factor of the thermoelement [4, 5]. Despite successes in the field of increasing the metallization adhesion for laboratory samples of single-crystal [2] and nanocrystalline [3] bismuth chalcogenides, the problem of low metallization adhesion to thermoelectric generator materials obtained by powder metallurgy methods remains urgent, and its solution is given insufficient attention. In scientific publications, there are practically no systematic studies of the dependence of the value of adhesion on the purity, structure, morphology, or surface roughness of such materials. In solving the problem of increasing adhesion, attention should be paid to the potential for hardening the surface layers of semiconductor legs by means of surface modification. In this aspect, tasks arise: selection of the optimal composition of the barrier layer and the

method of its application, the search for new methods of thermomechanical and chemical treatment of the surface of semiconductor legs, and the development of alternative methods for modifying the surface of thermoelectric legs, for example, laser, photon, plasma-chemical, or electronic treatment. For example, the efficiency of the laser surface treatment of Bi_2Te_3 samples in "healing" cracks and pores in a surface layer formed by cutting pressed samples was shown [6, 7]. Earlier, we proved the principal possibility and efficiency of pulsed photon treatment (PPT) by the radiation of powerful xenon lamps to increase the hardness of the surface layers of thermoelectric legs [8].

Thermoelectric legs of bismuth chalcogenides, suitable for the fabrication of thermogenerator batteries, are obtained by extrusion [9, 10] and pressing [11, 12]. To achieve a high quality of thermoelectric devices based on such polycrystalline samples, it is necessary to achieve the maximum anisotropy of the electrophysical properties of the rhombohedral crystal lattice. This problem is solved by technological extrusion modes. In the process of pressing, a crystallographic texture is formed, in which crystallographic planes with a high reticular density of atoms and cleavage planes in neighboring crystallites are located in the bulk of the material, predominantly, in the normal orientation to the pressing axis. Under hot pressing, material layers move in the surface layer in the sample-surface plane due to friction against the walls of the matrix, and texture is formed, characterized by the arrangement of cleavage planes along the sample surface. Such a texture reduces the mechanical properties of a thermoelectric leg, because microcracks form along the cleavage planes and sample lamination. The efficiency (reliability, durability, usefulness) of a thermoelectric generator (besides the material quality factor) greatly depends on the value of the contact resistance and adhesion at the semiconductor–metal interface [13, 14]. The latter, in turn, is determined by the properties of the material, the structure of the switching and antidiffusion layers, the parameters of the physicochemical interaction at the interface between the bismuth-chalcogenide solid solution–switching contact, and the conditions for modifying the semiconductor surface before applying metal layers.

The goal of the work is to determine the effectiveness of the photon treatment of semiconductor thermoelectric legs of *n*-type Bi_2Te_3–Bi_2Se_3 solid solution with incoherent light to improve their hardness and increase adhesion of the barrier and switching layers.

EXPERIMENTAL

Semiconductor branches based on the Bi_2Te_3–Bi_2Se_3 solid solution were obtained by the isostatic pressing of powders of the corresponding composition under argon at 650 K. After pressing, all the samples were polished using an emery-paper disk with a silicon-carbide abrasive grain from R2000 to R5000 to obtain a mirror surface. After mechanical treatment, the surface was cleaned of impurities by means of ultrasonic dispersion in distilled water.

Pulsed photon treatment (PPT) of the samples was performed by double irradiation (2×0.8 s) with packets of single pulses of 10^{-2} s each under argon, which corresponded to a radiation energy delivered to the sample of $E_I \sim 125$ J/cm^2. The mode was selected on the basis of previous studies on the effect of the PPT modes on the morphology, phase composition, and mechanical properties of the surface layers of *n*-type Bi_2Te_3–Bi_2Se_3 semiconductor samples [8].

Nickel films were used as the switching layer, and molybdenum films were used as the barrier layer. Metallic molybdenum and nickel were deposited layer-by-layer on the prepared surfaces of the semiconductor legs using a UVN-74M facility by magnetron sputtering of the corresponding targets in argon (4.2×10^{-4} mmHg). The samples were heated during the magnetron-sputtering process up to 470 K using IR (infrared) lamps. The power of the magnetron with a molybdenum target was 900 W, and with a nickel target, was 600 W; the growth rate of the molybdenum layer under these parameters was 3.3 nm/s, and for the nickel layer, it was 1.8 nm/s. The thickness of the metal layer was 270 nm for molybdenum and 250 nm for nickel.

To determine the effects of the mutual diffusion of atoms of the switching barrier layers and *n*-type semiconductor branches, all Bi_2Te_3–Bi_2Se_3/Mo/Ni heterostructures were subjected to isothermal annealing in evacuated (0.13 Pa) flasks at 570 K for 24 h.

The phase composition of the samples was studied by X-ray diffractometry (Bruker D2 Phaser). The elemental composition of the surface of the samples was investigated by local X-ray microanalysis using a JEOL JSM6380LV scanning electron microscope with a unit for EMF analysis (Oxford Instruments). The surface relief and roughness of the samples were studied by atomic force microscopy (NT-MDT Solver P47) and scanning electron microscopy (JEOL JSM6380LV and Quanta 600 3D). The hardness of the samples was investigated by measuring nanoindentation using a Nano Hardness Tester (CSM Instruments) with a Berkovich diamond indenter. The maximum load on the indenter was 10.0 and 200.0 mN. The loading rate during indentation was 15 and 300 mN/min, respectively. Each sample was measured at least eight times. The Meir hardness values of the surface layer and Young's modulus of the samples were determined by the Oliver–Pharr method according to *GOST* (State Standard) *R 8.748-2011*. The adhesion of the coatings was estimated using an Instron 5982 tensile shear test machine.

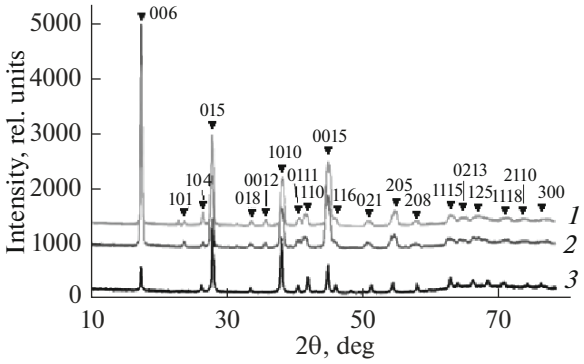


Fig. 1. X-Ray diffraction patterns of the legs based on $Bi_2Te-Bi_2Se_3$ solid solution (*1*) before and (*2*) after mechanical polishing and (*3*) after mechanical polishing + PPT.

RESULTS AND DISCUSSION

Morphology, phase composition, and mechanical properties of the surface of semiconductor legs after processing. *Phase composition*. Figure 1 shows the X-ray diffraction patterns characterizing the phase composition of surface layers of semiconductor legs before and after different surface-treatment options.

The diffraction pattern of the initial branch (curve *1*) contains reflections corresponding to the Bi_2Te_2Se phase with a rhombohedral lattice ($R\bar{3}m$) [15]. The relatively high intensity of the (006) and (0015) reflections of Bi_2Te_2Se indicates the presence of a preferred texture with the zone axis ⟨0001⟩.

As seen in Fig. 1, the phase composition of the surface layers of the semiconductor legs after mechanical polishing, and PPT did not change significantly. A decrease in the relative intensity of the (006) and (0015) reflections of Bi_2Te_2Se is observed both after mechanical polishing and after PPT. The samples after mechanical polishing exhibit broadened diffraction peaks, which indicates the stress that appears during deformation of the surface layer. After the PPT, the peak width coincides, as a rule, with the corresponding value for the sample prior to mechanical polishing. Since the used PPT modes result in the appearance of an effective temperature of ~800 K in the surface layer of the branches [8], it is clear that a decrease in the diffraction maximum width is due to recrystallization of the grains, which leads to a decrease in the stress in the surface layers. The relative decrease in the intensity of the (006) and (0015) reflections of Bi_2Te_2Se can also be a consequence of recrystallization.

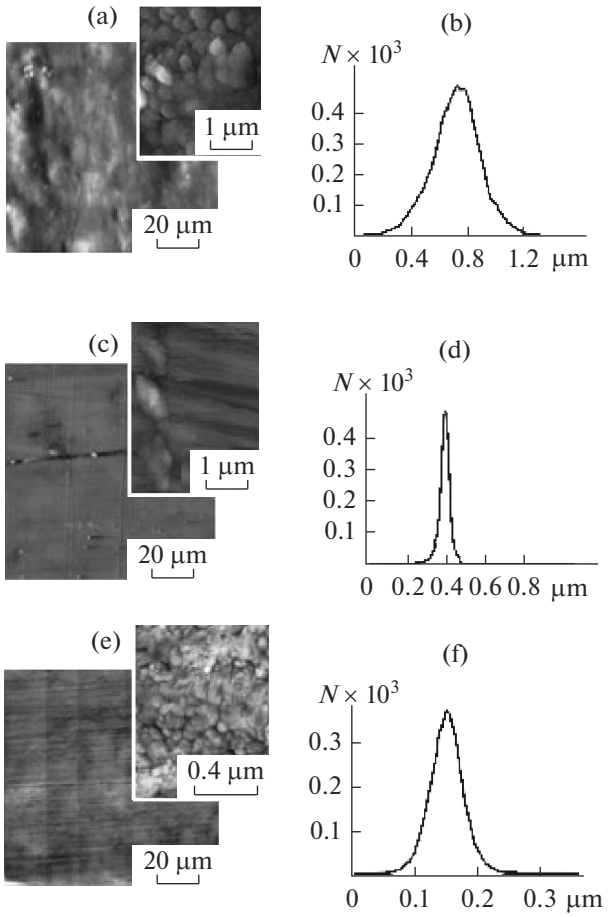


Fig. 2. AFM images and histograms of the height distribution of the surface topography of the bismuth-telluride leg (a, b) after pressing, (c, d) mechanical polishing, and (e, f) mechanical polishing + PPT.

Surface morphology. The AFM images and histograms of the height distribution for the initial samples and the samples after processing are presented in Fig. 2. The data on the analysis of the surface relief are given in Table 1.

The initial sample surface has a bimodal pattern of relief inhomogeneities (Fig. 2). First, these are inhomogeneities with sizes substantially larger than 3 μm (both in the lateral direction and in height) and, second, inhomogeneities up to 1 μm in the plane and no

Table 1. Surface-relief parameters of the semiconductor legs of bismuth telluride before and after processing

Surface treatment	Height variance, nm	Roughness over an area of 0.01 mm^2, nm
Initial sample	1508	137
Mechanical polishing	1074	23
Mechanical polishing + PPT	363	22

Table 2. Hardness and modulus of elasticity of the surface layer of a semiconductor leg that has undergone various types of treatment

Type of treatment	$F = 10$ mN		$F = 200$ mN	
	H, GPa	E, GPa	H, GPa	E, GPa
Without treatment	1.1	29.7	0.9	29.5
Mechanical polishing	1.3	26.9	0.7	21.6
Mechanical polishing + PPT	1.3	31.2	0.9	32.8

Table 3. Adhesion of coatings and nature of destruction of the Bi_2Te_2Se/Mo/Ni heterostructure

Type of treatment	R_{shear}, MPa	Nature of destruction
Without treatment	1.3	Cohesion
Mechanical polishing	5.6	Combined
Mechanical polishing + PPT	5.5	Adhesion

larger than 0.3 μm along the vertical direction. Apparently, defects of the former type are due to dispersion of the initial powder of the $Bi_2Te-Bi_2Se_3$ solid solution and the crystallite morphology formed during isostatic pressing.

Defects of the latter type are the places of stress concentration as a result of plastic deformation (extrusion), leading to a change in the geometry and size of crystallites and surface structure and roughness. Figure 2a shows an AFM image of a region that contains only defects of the latter type. The surface morphology of the $Bi_2Te_3-Bi_2Se_3$ legs inherits the irregularities (roughness) of the mold.

The asymmetry of the histogram of the relief height distribution (Fig. 2b) indicates that depressions prevail over protrusions; that is, no defects caused by material extrusion along the cleavage planes were revealed.

After mechanical polishing, the surface of the $Bi_2Te-Bi_2Se_3$ leg has a nonuniform relief (Fig. 2c); the height difference does not exceed 0.15 μm, which is substantially smaller than the grain sizes of the P2000 and P5000 abrasives used in polishing. The asymmetry of the histogram indicates the dominance of the depressions over protrusions, which indicates that polishing leads to the elimination of defects protruding above the surface of the material. Mechanical polishing artifacts are scratches and small fragments on the surface of the leg, which are successfully removed in the process of ultrasonic treatment.

After PPT, there is a change in the nature of the relief: a homogeneous relief is formed (Fig. 2e), containing defects of the same type, the morphology of which indicates the possible nature of their appearance, that is, the recrystallization of crystallites and changes in the structure of the surface layer of the semiconductor. The height difference does not exceed 0.1 μm; the histogram (Fig. 2f) reflects the normal distribution of inhomogeneities in height, which indicates the uniform crystalline structure of the surface layer of the material.

Mechanical properties. The hardness H and the modulus of elasticity E of the surface layer of the semiconductor legs after processing are given in Table 2.

It is known that the hardness of highly dispersed coatings made of solid solutions based on bismuth telluride is up to 2.5 times higher than that of single crystals of the corresponding composition [16]. The Young's modulus in coarse-grained (grain size of approximately 20 μm) bismuth telluride with a uniaxial texture [0001] is minimal along the [0001] direction and is about 32 GPa, while with a high dispersion of the grain structure (average grain size of approximately 0.2 μm) and a free grain orientation, Young's modulus of bismuth telluride is about 48 GPa [17].

Mechanical polishing results in an increase in hardness and Young's modulus of the surface layer of the legs under a load of 10 mN and their decrease under a load of 200 mN (Table 2). The observed result can be attributed to removal of the damaged layer in the process of mechanical polishing, located in the surface region of the original legs; this layer differs from the inner regions in both chemical composition and a lack of texture [0001]. A relatively thin deformed layer having a dispersed grain structure is formed instead of the removed layer. The values of H and E obtained at a low load applied to the indenter (10 mN) characterize only the surface layer, and at a load of 200 mN in the case of the polished leg, inner layers with a coarse grain structure and texture [0001] contribute to the values of H and E.

The PPT of mechanically polished legs leads to the formation of surface layers with an arbitrary grain orientation and a relatively large thickness, compared with the depth of penetration of the indenter. The consequence of this is E values, close to both indentation loads and increased compared to the mechanically

polished legs, and the H value increased under a load of 200 mN.

Adhesion properties of the surface layer of the legs. X-ray diffraction study of the $Bi_2Te-Bi_2Se_3$/Mo/Ni heterostructures showed that the phase composition is the same for all types of surface treatment of the semiconductor samples. The characteristic fragment of the X-ray diffraction pattern from the semiconductor sample after mechanical polishing, ultrasonic treatment, and PPT with applied switching and barrier layers of nickel and molybdenum is presented in Fig. 3.

The analysis showed that the switching, barrier, and surface layers of the semiconductor sample contain only pure metallic phases of molybdenum and nickel and the Bi_2Te_2Se phase. The data obtained indicate that when using the magnetron sputtering of molybdenum and nickel on the semiconductor surface and their condensation at a temperature of 470 K, chemical interaction of the components does not occur at the interfaces of the Bi_2Te_2Se/Mo/Ni heterostructure nor do new crystalline phases form.

Using shear tests, it was found that only the Bi_2Te_2Se phase was detected on the heterostructure surface in the metallization-separation region, which indicates destruction of the heterostructure along the interface (adhesive failure) or semiconductor volume (cohesive failure). Table 3 shows the adhesion values of the coatings and the nature of destruction of the Bi_2Te_2Se/Mo/Ni heterostructure, found based on the results of AFM study of the destroyed surface (detached) from the coating side.

For coatings obtained on the unmodified (initial) surface of the legs, the adhesion value is the smallest, which is due to contamination and a high density of bulk defects (cracks, pores) in the surface layer. The adhesion of coatings on the surface of the legs after mechanical polishing, as well as mechanical polishing and PPT, is more than four times the adhesion of the coating on the original leg. An increase in the adhesion value correlates with the nanoindentation data, which may be due to the presence of a hardened layer in the surface region of the semiconductor with a highly dispersed grain structure.

Mutual diffusion of atoms of the $Bi_2Te_3-Bi_2Se_3$/Mo/Ni heterostructure. We studied the surface of $Bi_2Te_3-Bi_2Se_3$/Mo/Ni heterostructures that underwent isothermal annealing in vacuum at 570 K for 24 h using scanning electron microscopy (SEM) and electron probe microanalysis. Heterogeneous regions with lateral dimensions of 40 to 250 nm were found as a result, which contain atoms of selenium and tellurium.

The SEM images and profiles of the elemental composition after isothermal annealing of $Bi_2Te_3-Bi_2Se_3$/Mo/Ni heterostructures created by different methods of modifying the surface of the legs are presented in Fig. 4.

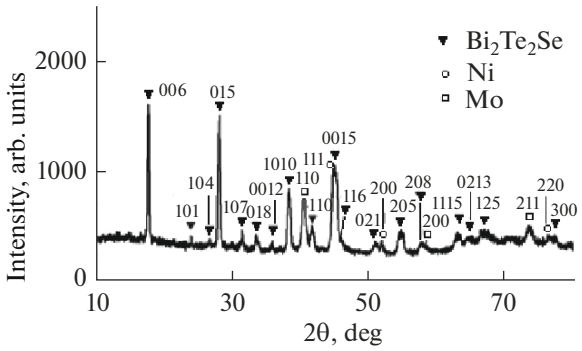


Fig. 3. X-Ray diffraction pattern from the semiconductor leg after mechanical polishing, and PPT with applied layers of metallic nickel and molybdenum.

The structural resolution enables us to distinguish regions and interphase boundaries separating thin metallization layers and a massive $Bi_2Te_3-Bi_2Se_3$ semiconductor solid solution in the cross section of the heterostructure obtained by the vacuum condensation of metals (molybdenum and nickel) onto the surface of a leg that was previously mechanically polished (Fig. 4a). The semiconductor is characterized by a large-block crystal structure with crystallite sizes from 0.3 to 3.0 μm. Some porosity and the formation of cracks are observed, as a rule, along the grain boundaries in the surface region of the $Bi_2Te-Bi_2Se_3$ solid-solution leg. The gradient of the elemental composition in the surface region of the $Bi_2Te-Bi_2Se_3$/Mo/Ni heterostructure is illustrated in Fig. 4c. Mo/Ni metallization with a thickness of each layer of ~250 nm completely covers the pores and cracks in the surface layer of the mechanically polished $Bi_2Te-Bi_2Se_3$ solid solution. The top layer of nickel metallization has a highly dispersed grain structure with an average grain size of ~50 nm and closed pores of the same size. Since the nickel layer after deposition did not contain pores, their formation is due to diffusion and recrystallization processes taking place during isothermal annealing in the surface metallization layer at the nickel–tellurium interface, which is formed during desublimation. Desublimation is an exothermic phase transition; therefore, the effective temperature can be increased and a physicochemical interaction at the interface can be activated. The bottom molybdenum layer is continuous (without pores); the grain structure cannot be resolved by SEM, since it is nanocrystalline.

Investigation of the elemental composition showed that mutual diffusion of the elements of the $Bi_2Te-Bi_2Se_3$/Mo/Ni heterostructure during annealing (at 300°C for 24 h) does not lead to a significant change in the elemental composition of the individual layers of the heterostructure. At the same time, a high concentration of selenium and tellurium is observed in the metallized surface of the heterostructure, which is caused by the mass transfer of tellurium and selenium

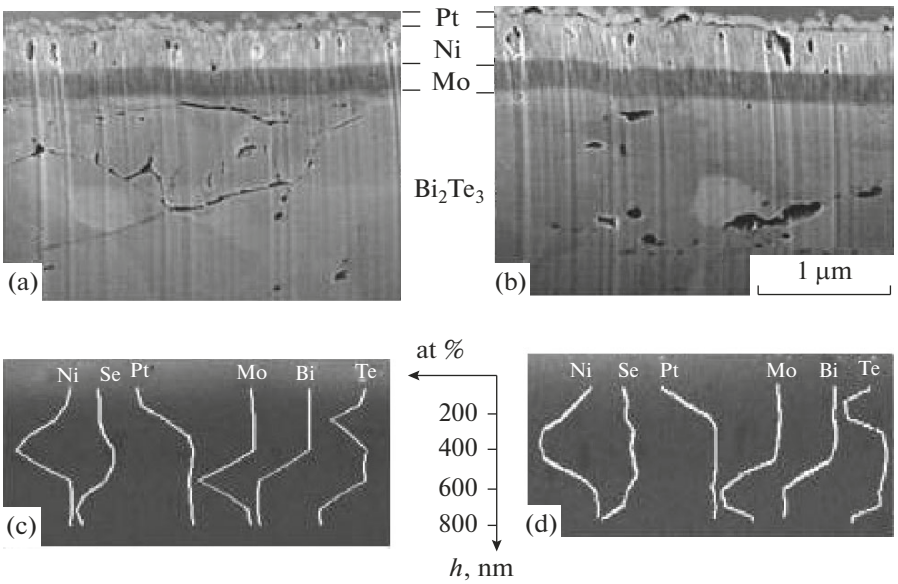


Fig. 4. (a, b) SEM images and (c, d) elemental composition of the profile of the $Bi_2Te-Bi_2Se_3$/Mo/Ni heterostructure (mechanically polished $Bi_2Te_3Bi_2Se_3$/Mo/Ni solid solution) after heat treatment at $T = 500$ K in for 24 h after (a) mechanical polishing and (b) mechanical polishing + PPT.

in the processes of sublimation from the lateral surface and subsequent desublimation on the metallized surface rather than by the bulk diffusion of their atoms. The mechanism of formation in the metallization layer of inhomogeneous porous regions (up to 100 nm) is due to the mass transfer of nickel to the selenium-telluride layer desublimated on its surface. At a depth of approximately 600 nm from the outer surface of the heterostructure, the atomic concentration of bismuth and then selenium and tellurium reach values corresponding to the stoichiometry of the $Bi_2Te_{2.4}Se_{0.6}$ solid solution. Taking into account the metallization thickness (~500 nm), deviation of the atomic concentration of elements from stoichiometry in the bulk of the $Bi_2Te-Bi_2Se_3$ solid solution is observed at a thickness of no more than 100 nm. Thus, the application of a magnetron layer of molybdenum with a thickness of 250 nm to the $Bi_2Te-Bi_2Se_3$ solid-solution surface is a sufficient condition for preventing the interdiffusion of elements in the heterostructure of the mechanically polished $Bi_2Te-Bi_2Se_3$/Mo/Ni solid solution after annealing at $T = 570$ K for 24 h.

A SEM image of the cross section of the heterostructure created by the vacuum condensation of metals (molybdenum and nickel) onto the surface of the mechanically polished and PPT-modified branch shows well resolved separate metallization layers and the crystal structure of the $Bi_2Te_3-Bi_2Se_3$ solid solution (Fig. 4b). As before, pores and cracks dominate in the surface region of the $Bi_2Te_3-Bi_2Se_3$ solid-solution leg. However, in contrast to mechanical polishing, the fraction of cracks and open pores in the semiconductor surface layer is significantly smaller after PPT, which is caused by the energy effect of the PPT and local recrystallization of the imperfect surface layer of the solid solution at a depth of 100–200 nm.

The gradient of the elemental composition in the surface region of the $Bi_2Te-Bi_2Se_3$/Mo/Ni heterostructure (Fig. 4d) is similar to that discussed above for the structure obtained by preparing the surface of the legs with mechanical polishing only. The mutual diffusion of atoms of the metallization (nickel and molybdenum) and the $Bi_2Te-Bi_2Se_3$ solid solution during isothermal annealing at 570 K is practically lacking; the proportion of nickel and molybdenum at a depth of ~1 μm from the surface of the solid-solution leg does not exceed 0.1 at %.

CONCLUSIONS

Preparing the surface of thermoelectric legs based on the *n*-type solid solution $Bi_2Te-Bi_2Se_3$ by mechanical polishing of the surface followed by pulsed photon treatment leads to a three- to fourfold increase in the adhesion of the barrier and switching Mo/Ni layers.

The energy effect of the PPT stimulates local recrystallization of the imperfect layer near the surface of the semiconductor legs at a depth of 100–200 nm, increasing the hardness of the surface layers of the $Bi_2Te-Bi_2Se_3$ solid solution.

It is shown that the proposed method including PPT and modification of the surface of the thermoelectric branches practically excludes the mutual diffusion of atoms of the switching layers (nickel and molybdenum) and the $Bi_2Te-Bi_2Se_3$ solid solution during isothermal annealing at a temperature of 570 K.

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